

TANMAI VEERAPANENI

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EDUCATION

Master of Science in Engineering Data Science | University of Houston May 2025
Houston, Texas, 77054 | **GPA: 3.96/4**
Bachelor of Technology in Computer Science | Velagapudi Ramakrishna Siddhartha Engineering College May 2023
Vijayawada, India, 520007 | **GPA: 9.15/10**

EXPERIENCE

Research and Development Intern, Navtech Electronics Pvt. Ltd. June 2022-September 2022

- Designed a scalable cloud architecture on AWS for real-time locator systems, using EC2, S3 and Kinesis.

Robotic Process Automation Internship, Blue Prism March 2022-May 2022

- Implemented a project titled "Digital Worker for Grading System" with a performance efficiency of 87%.

Research Assistant, ISRO Project, Velagapudi Ramakrishna Siddhartha Engineering College June 2022-July 2023

- Worked on the project with the "Indian Space Research Organization" & "North Eastern Space Research Organization".

TECHINICAL SKILLS

Programming Languages: Python (PySpark, Pandas, TensorFlow, Scikit-learn, Numpy, Matplotlib), Java, R, SQL.
Tools and Technologies: Databricks, Snowflake, Power BI, Tableau, Hadoop, Airflow, Kafka, Excel, Git.
Core Concepts: Data Structures, Data Engineering & Analytics, Machine Learning, Big Data, Gen AI, Agentic AI, CI/CD.
Cloud Technologies: Amazon Web Services, Microsoft Azure.

PROJECTS

Spotify Data Pipeline Using AWS May 2025- Present

- Tech stack:** Snowflake, Spark, AWS Lambda, AWS S3, AWS Glue, AWS Athena
- Engineered a Spotify data pipeline and optimized data loading by 60% and query performance by 50%.
- Currently enhancing the pipeline's transformation layer using AWS Glue ETL and Apache Spark.

E-commerce Data Pipeline with SCD Modelling using Medallion Architecture February 2025- April 2025

- Tech stack:** AWS S3, Databricks, SQL
- Built an end-to-end DLT ETL pipeline on Databricks with Type1/2 modeling using medallion architecture for data integrity.
- Automated the pipeline with Databricks workflows, leveraging Unity Catalog for governance & real-time alerts for reliability.

Visual Analytics Using FrostByte databases for review analysis November 2024- January 2025

- Tech stack:** Snowflake, AWS S3, Tableau, PostgreSQL
- Extracted data & warehoused in Snowflake's data platform by integrating with AWS cloud for data pipeline.
- Developed interactive dashboards in Tableau for visual insights that reduced the reporting time by 30%.

Quantum Computing in Image Processing January 2024-April 2024

- Tech stack:** Python, ML, CUDA with Python, Qiskit, PennyLane
- Explored the intersection of quantum computing and digital image processing using superposition and entanglement.
- Demonstrated the potential of quantum algorithms with 80% better results using Qiskit and PennyLane.

Bamboo Mapping Using Supervised Learning, GEE, QGIS and Hybrid U-Net July 2022-April 2023

- Tech stack:** Python, Google Earth Engine, QGIS, Keras, TensorFlow
- Funded by ISRO and NESAC & masked satellite image is given as input & pre-processing is done.
- Hybrid U-Net model is trained & tested with 93% overall accuracy, with labels of different types of bamboo for classification.

CERTIFICATIONS & PATENT

Tableau Desktop Specialist April 2025
Microsoft Certified: Azure Data Engineer Associate May 2024
British Airways – Data Science Job Simulation Virtual Internship May 2024
Introduction to R Software by NPTEL (By the Government of India) November 2020
Patent, Intelligent Smart Cooker July 2023

POSITIONS OF RESPONSIBILITY

Cultural Volunteer, GISO, University of Houston August 2023-December 2024
Finance Head, MAD (Make A Difference), NGO July 2021-January 2023